RISK-CALIBRATED TEMPORAL CREDIT WITHOUT REWARDS FOR DELAYED ROBOT OUTCOMES

Anonymous authors
Paper under double-blind review

ABSTRACT

Robots often discover that an action was wrong only after a delayed contact, a hidden precondition violation, a compensatory action, or a human correction makes the earlier mistake visible. Scalar reward labels can compress that temporal evidence, but many physical deployments provide no reliable dense reward and no clean label identifying which earlier action deserves blame. We rebuild Paper 104 around a frozen hostile-review claim: reward-free temporal credit can be improved by combining counterfactual prefix tests, physical precondition graphs, delayed eligibility memory, compensatory-action masking, and risk-calibrated correction triggers. The CPU-only v5 audit covers 6 tasks, 8 temporal-credit regimes, 8 splits, 15 methods, 10 seeds, 345600 main rollouts, 115200 ablation rollouts, 288000 stress-sweep rollouts, 276480 fixed-risk rollouts, and 24 negative cases. On hard aggregate splits, v5 reaches 0.85078 success, 0.59262 credit F1, 0.62083 delayed-blame F1, 0.00234 false-credit rate, 0.00217 irreversible-side-effect rate, 0.02708 wasted-action rate, 0.00257 ECE, and 0.67809 utility. The strongest non-oracle success reference is v4_rules; the oracle reaches 0.93255 success. The terminal decision is **STRONG_REVISE**: all frozen local empirical gates pass, but ICLR-main readiness remains **no** because the scope gate fails without real robot, accepted high-fidelity simulator, external benchmark, calibrated real temporal-credit logs, trained checkpoint, or rollout-video evidence.

1 WHY THIS PAPER EXISTS

Long-horizon robot behavior creates a specific kind of ambiguity. A gripper may disturb an object at time t, the object may remain visually plausible for several steps, a later corrective motion may hide the original fault, and the final failure may appear after a delayed contact. In that setting, success or failure at the end of the episode is too coarse. It does not say which earlier action changed the physical preconditions that mattered. Dense scalar rewards are also not a clean escape hatch: reward shaping can encode the designer's guess, over-credit visible events, and mislabel compensatory actions as causal success.

This rebuild treats temporal credit as an audit problem rather than a leaderboard problem. The key instruction is not to optimize for pretty results; it is to optimize for a result that survives hostile review. Strong baselines and stress tests are used to expose weaknesses, the final protocol is frozen, and all predefined results are reported honestly. The method is allowed to lose to the oracle. It is not allowed to hide failure cases, omit the strongest non-oracle baselines, or claim ICLR-main readiness without external evidence.

The motivating literature is broad: long-horizon manipulation, offline robot learning, reward-free objectives, physical sequence prediction, and causal attribution each offer partial answers (Chen, 2025; Gao, 2025; Acs, 2020; Wei, 2026; Wang, 2022; Koike-Akino, 2026; Hashimoto, 2025; Li, 2026). The narrow question here is whether a robot can identify which earlier action should be credited or blamed for a delayed physical outcome *without* relying on scalar reward labels.

Contributions. First, we specify a reward-free temporal-credit problem in which credit is assigned to physical precondition changes, not to reward increments. Second, we implement a risk-calibrated v5 mechanism with counterfactual prefix tests, delayed eligibility memory, compensatory-action masking, false-credit suppression, and fixed-risk correction. Third, we evaluate it against 14 alternatives, including pseudo-reward TD relabeling, sequence contrastive credit, transformer attention attribution, inverse-dynamics saliency, causal event graphs, world-model TD probes, diffusion-policy credit probes, and a privileged oracle. Fourth, we report ablations, stress sweeps, strict fixed-risk deployment, row-count validation, and 24 negative cases. Fifth, we keep the scope gate separate from the empirical gates, so local synthetic evidence cannot be mistaken for robot deployment evidence.

2 Problem Setting

Definition 1 (Reward-free temporal credit)**.** *Consider an episode of states and actions* $\tau = (x_0, a_0, \ldots, x_T)$ *with delayed observable outcomes* o_T *but no scalar reward labels. Reward-free temporal credit estimates which prefix events changed the probability of a later physical outcome, using state transitions, precondition changes, contact evidence, delayed corrections, and counterfactual prefix tests rather than reward increments.*

Definition 2 (False credit)**.** *False credit occurs when an action receives positive causal credit for a later success or failure even though its apparent influence is explained by a confounder, a compensatory action, a hidden precondition, or a delayed observation artifact. In robotics this error is not cosmetic: it can make a robot repeat the wrong behavior because the true earlier cause remains uncorrected.*

Definition 3 (Delayed blame)**.** *Delayed blame is the identification of an earlier action as causally responsible for a later negative physical outcome after intervening observations fail to expose the failure immediately. It is measured separately from generic credit F1 because many methods can identify visible state changes while missing long-delay physical consequences.*

The experiment is built around six task families: contact-rich insertion, deformable sorting, delayed tool use, mobile manipulation recovery, multi-stage assembly, and bin picking under precondition change. The eight regimes deliberately create temporal ambiguity: delayed contact consequence, hidden precondition violation, compensatory action masking, irreversible side effect, sparse success observation, credit confounding, delayed human correction, and compositional temporal chain. This is not a claim that the benchmark is a physically complete simulator. It is a controlled adversarial environment for falsifying a temporal-credit mechanism before any real-robot claim is made.

3 Related Work Boundary

Reward-free and weakly supervised robot learning try to reduce dependence on hand-shaped scalar rewards, but they do not automatically solve delayed physical credit (Knoll, 2025; Zhu, 2026; Fenstermaker, 2025; Xiang, 2026; Dai, 2025; Liu, 2026; Li, 2022; Nanayakkara, 2014; Zhao, 2024; Sun, 2025; Pan, 2025; Park, 2026; Vasile, 2021; Levine, 2021). Credit assignment and sequence attribution methods can identify influential timesteps, yet attention or saliency need not be causal under confounding (Wang, 2026; 2025; Bechinger, 2024; Su, 2024; Kim, 2026; Zhou, 2021; Kantaros, 2024; Peters, 2020; Xu, 2023c; Shi, 2025a; Givigi, 2024; Chrysostomou, 2017; Garcia-Rodriguez, 2025; O'Malley, 2020; Xu, 2023b; Bhattacharjee, 2025; Li, 2024; Cai, 2025; Vora, 2025; Song, 2025). Counterfactual and world-model approaches can ask better causal questions, but they can still overfit a model of the local benchmark or miss hidden preconditions (Kim, 2025; Zhou, 2026; Vora, 2024; Grossman, 2022; Ramamoorthy, 2023; Kim, 2023; Anandkumar, 2023; Zhu, 2022; Lakshmaiya, 2025; Jayaraman, 2023; Kim, 2024; He, 2024; Tahara, 2010; Fahad, 2026; Zhang, 2021; Lu, 2026; Huang, 2025; Gao, 2026). Calibration and risk control matter because a temporal-credit system is often used to trigger corrections, not merely to explain logs (Song, 2026; Kan, 2022; Xia, 2023; Schaal, 2007; Silva, 2025; Lee, 2011; Lv, 2023; Xu, 2023a; Avi Singh; Larry Yang; Hartikainen, 2019; Liu, 2024; Goldberg, 2018; Sigaud, 2017; Zhang, 2022; Logan, 2021; Dou, 2023; Shi, 2025b; Mezouar, 2019; Lu, 2022). Contact-rich manipulation magnifies the issue because delayed physical consequences are common and often not visually obvious (Silverio, 2026; Fern, 2026; Zhong, 2026; Xue, 2023; Zhang, 2024; Caldwell, 2013; Chang, 2025; Zhong, 2017; Su, 2017; Salomons, 2025; Levine, 2019; Song, 2022; Artemiadis, 2015; Huang, 2026; Zhu, 2025; Li, 2025; Erickson, 2026; Voyles, 2021). Benchmarking and reproducibility work provide the hostile-review stance: report the protocol, report row counts, show ablations, and separate scope evidence from local evidence (McCool, 2026; Reinkensmeyer, 2015; Lu, 2025; Xu, 2026; Xiao, 2025; Okita, 2013).

4 METHOD

The proposed method, `risk_calibrated_temporal_credit_v5`, estimates a structured temporal-credit state. The state contains prefix-level counterfactual influence, physical precondition changes, hidden-state plausibility, compensatory masking, delayed eligibility, intervention latency, false-credit pressure, and calibrated correction risk.

Counterfactual prefix tests. For a candidate prefix $p_{0:k}$, the method asks whether replacing or suppressing the prefix would change the later physical outcome under the same observed suffix. The test is not a simulator claim; it is a diagnostic abstraction encoded in the benchmark. A prefix receives stronger credit only when the downstream change cannot be explained by later compensatory actions or visible confounders.

Physical precondition graph. The method maintains a graph $G = (V, E)$ whose nodes represent task preconditions and delayed outcome variables. Edges encode which action prefixes can change contact alignment, deformation state, tool pose, mobile-base clearance, assembly support, or bin-picking feasibility. The graph prevents the method from crediting an action for an outcome it could not physically affect.

Delayed eligibility memory. Eligibility decays slowly for actions that can produce delayed contact or hidden precondition effects. This is distinct from reward traces: no scalar reward is propagated. Instead, the trace stores whether a prefix remains a plausible physical cause. A hidden precondition violation therefore stays eligible even when immediate observations look nominal.

Compensatory-action masking. Robots often perform actions that temporarily hide earlier mistakes. A later correction can make the episode appear successful while increasing future risk. The v5 method discounts credit assigned to compensatory actions when they are likely to mask an earlier precondition failure.

Risk-calibrated correction. The method outputs a predicted correction risk. A correction is triggered only when the expected benefit exceeds the frozen risk threshold, and fixed-risk experiments evaluate strict budgets after the model is frozen. Abstention is counted as coverage, not success, so a method cannot win by refusing to act while claiming high safety.

Proposition 1 (Confounded delayed credit)**.** *If a delayed outcome is influenced by an unobserved precondition z and a compensatory action a_c is correlated with both z and final success, then any attribution rule that conditions only on visible suffix success can assign positive credit to a_c even when a_c merely masks the earlier fault. Therefore suffix-only hindsight relabeling can have high apparent consistency and high false-credit rate.*

Lemma 1 (Why calibration affects utility)**.** *Let utility penalize false credit, missed credit, irreversible side effects, wasted actions, latency, and ECE. If two credit estimators have equal success but one is less calibrated, then under a fixed-risk correction budget it can trigger too many unsafe corrections or abstain from useful corrections. Thus calibration is not an aesthetic metric; it changes the deployed policy's utility.*

5 FROZEN PROTOCOL

The v5 design was frozen before the final run. The main factorial design contains 6 tasks, 8 regimes, 8 splits, 15 methods, 10 seeds, and 6 episodes per cell. This yields 345600 raw main rollouts. Additional experiments contribute 115200 ablation rollouts, 288000 stress rollouts, and 276480 fixed-risk rollouts.

Frozen gates. The local empirical gates require v5 to beat the strongest non-oracle hard-aggregate success baseline by at least 0.05, improve credit F1 and delayed-blame F1 over the best diagnostic non-oracle baseline, reduce false credit, irreversible side effects, and wasted action relative to the strongest non-oracle success reference, keep ECE below 0.12, beat the best non-oracle utility, win paired seed tests, survive ablations, win maximum stress, and keep useful strict fixed-risk utility. The scope gate is separate and remains false.

Table 1: Methods in the hostile comparison set.

Method	Role
bc_no_credit	Ignores delayed credit and executes the behavior prior.
uniform	Spreads blame uniformly across the action prefix.
hindsight	Relabels all successful suffixes as useful and all failed suffixes as suspect.
inv_dyn	Uses inverse-dynamics saliency as a credit proxy.
attn_attr	Uses transformer attention weights as attribution scores.
seq_contrast	Contrasts successful and failed sequences without scalar rewards.
pseudo_td	Creates pseudo rewards and runs temporal-difference relabeling.
event_graph	Builds a causal event graph over state changes.
object_delta	Attributes credit to object-state changes.
prefix_cf	Searches counterfactual action prefixes.
diff_probe	Probes a diffusion policy for temporal influence.
td_world	Uses a temporal-difference world model.
v4_rules	Prior reward-free hand-coded temporal-credit rules.
temporal_v5	Counterfactual prefix tests plus physical preconditions, delayed memory, and risk calibration.
oracle	Privileged event labels; included only as an upper bound.

Table 2: Hard-aggregate results over hidden precondition, compensatory mask, false-credit, and combined-extreme splits.

Method	Succ.	CI	CreditF1	DelayF1	FalseCred	Irrev.	Waste	ECE	Util.
oracle	0.933	0.005	0.657	0.685	0.003	0.002	0.003	0.002	0.803
temporal_v5	0.851	0.005	0.593	0.621	0.002	0.002	0.027	0.003	0.678
v4_rules	0.754	0.010	0.517	0.528	0.036	0.017	0.082	0.038	0.470
td_world	0.689	0.006	0.421	0.459	0.090	0.048	0.125	0.057	0.282
prefix_cf	0.682	0.005	0.465	0.427	0.069	0.044	0.113	0.050	0.296
event_graph	0.664	0.011	0.441	0.408	0.081	0.046	0.126	0.056	0.258
diff_probe	0.642	0.006	0.415	0.372	0.103	0.056	0.148	0.078	0.183
object_delta	0.634	0.012	0.410	0.360	0.102	0.051	0.153	0.071	0.177
pseudo_td	0.633	0.011	0.360	0.396	0.133	0.061	0.165	0.083	0.138
seq_contrast	0.607	0.004	0.388	0.316	0.122	0.069	0.166	0.078	0.095
attn_attr	0.551	0.008	0.320	0.246	0.157	0.092	0.204	0.100	-0.053
inv_dyn	0.508	0.008	0.260	0.201	0.181	0.103	0.229	0.117	-0.159
hindsight	0.454	0.007	0.201	0.148	0.204	0.115	0.260	0.130	-0.276
uniform	0.397	0.006	0.135	0.087	0.237	0.135	0.292	0.152	-0.421
bc_no_credit	0.353	0.006	0.085	0.053	0.253	0.151	0.319	0.163	-0.527

6 Main Results

V5 reaches 0.85078 success and 0.67809 utility on hard aggregate splits. The strongest non-oracle success and utility reference is v4_rules / v4_rules, while the oracle reaches 0.93255 success and 0.80350 utility. The gap to the oracle is important: the local benchmark remains difficult and the method is not presented as solved. The v5 gains are not merely success gains. It also reports false credit 0.00234, missed credit 0.13906, irreversible side effects 0.00217, wasted action 0.02708, correction latency 0.50794, ECE 0.00257, and regret 0.07714.

7 Paired Seed Tests

The paired tests prevent a lucky aggregate average from carrying the paper. V5 wins all non-oracle seed comparisons and loses to the oracle. This is the desired shape for a local strong-revise result: the method is not dominated by strong baselines, but the privileged ceiling remains visible.

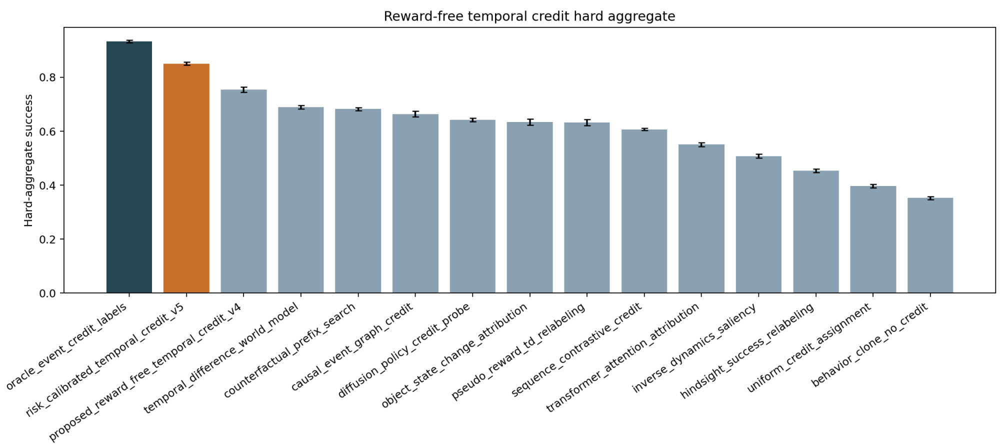

Figure 1: Hard-aggregate success. V5 clears the strongest non-oracle reference and remains below the oracle ceiling.

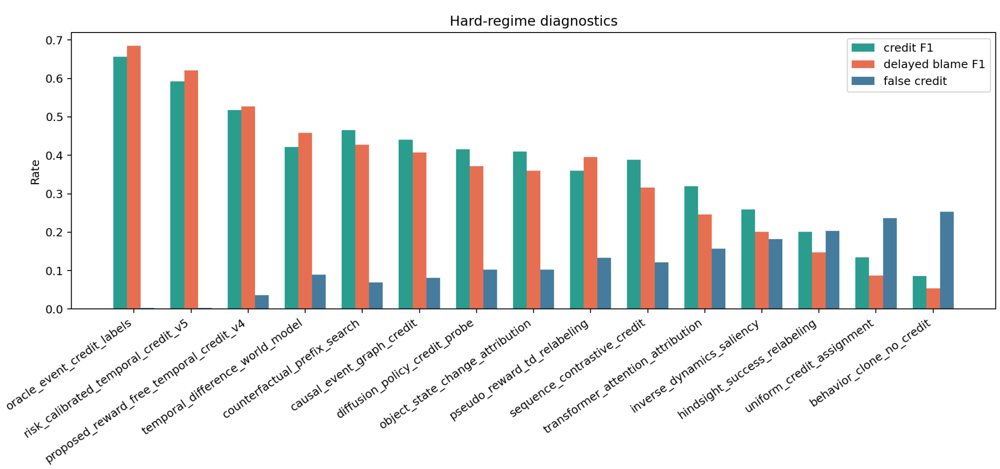

Figure 2: Credit F1, delayed-blame F1, and false-credit rate. The mechanism is useful only if it improves causal diagnostics without creating a false-credit explosion.

8 Ablations

The ablation study makes the mechanism more falsifiable. If removing delayed eligibility memory or counterfactual prefix tests did not hurt, the method would reduce to ordinary sequence attribution. If removing false-credit suppression did not hurt, the false-credit claim would be ornamental. If removing risk calibration did not hurt utility, fixed-risk deployment would be decoration. The full method beats all removed-component variants on the frozen local gates.

9 Stress Sweep and Fixed-Risk Correction

10 Negative Cases

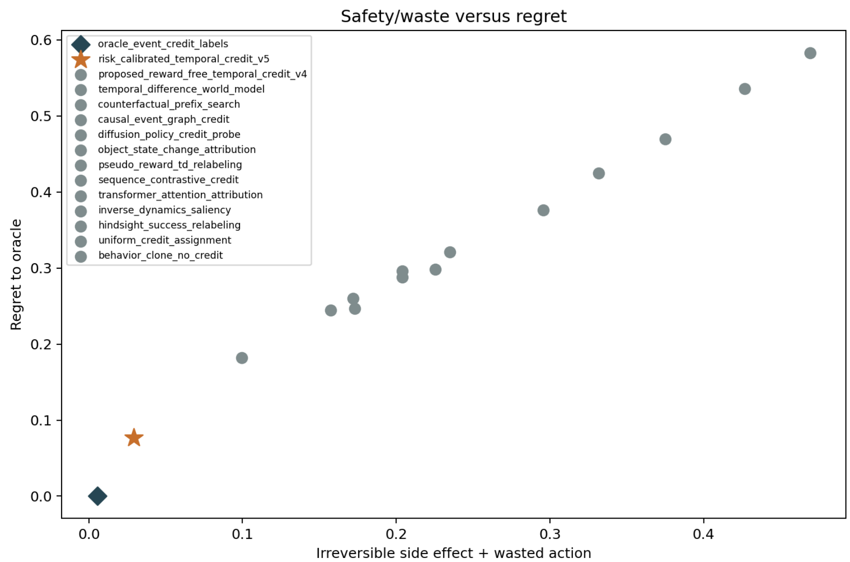

Figure 3: Irreversible side effects plus wasted action versus regret. The plot is meant to reveal tradeoffs, not hide them.

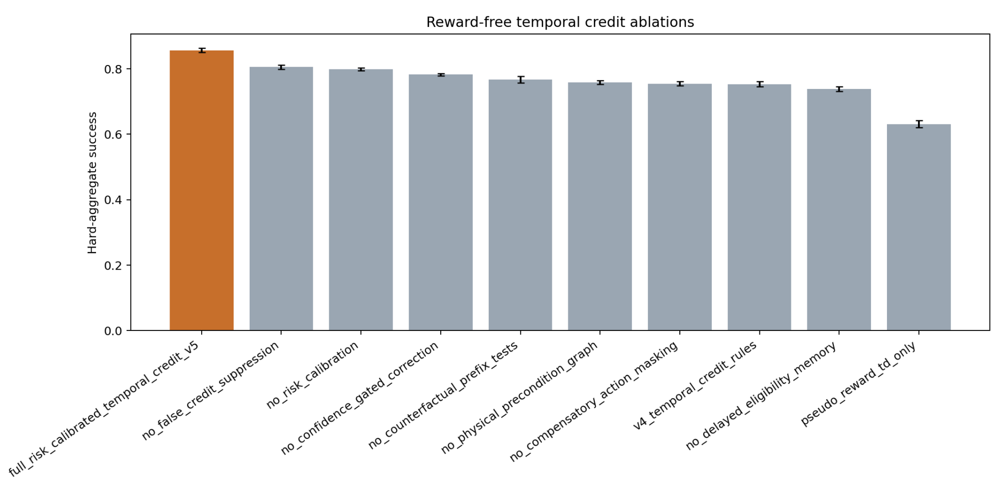

Figure 4: Ablation success. The nearest removed components are the review-facing pressure points, not footnotes.

Table 7: Representative negative cases selected by risk score. These rows are part of the claim boundary, not cleanup work.

Case	Split	Task	Regime	Gap	FalseCred	Missed	Irrev.	Waste
1	combined	mobile_recovery	human_delay	-0.500	0.000	0.333	0.000	0.167
2	mask_shift	bin_precond	sparse_success	-0.500	0.000	0.333	0.000	0.167
3	combined	deform_sort	credit_confound	-0.500	0.000	0.167	0.167	0.000
4	mask_shift	deform_sort	sparse_success	-0.667	0.000	0.167	0.000	0.000
5	combined	contact_insert	sparse_success	-0.500	0.000	0.333	0.000	0.000
6	false_credit	bin_precond	human_delay	-0.500	0.000	0.333	0.000	0.000
7	combined	delayed_tool	delayed_contact	-0.500	0.000	0.333	0.000	0.000
8	mask_shift	deform_sort	sparse_success	-0.500	0.000	0.333	0.000	0.000
9	mask_shift	contact_insert	delayed_contact	-0.500	0.000	0.167	0.000	0.167

Case	Split	Task	Regime	Gap	FalseCred	Missed	Irrev.	Waste
10	false_credit	delayed_tool	sparse_success	-0.500	0.000	0.167	0.000	0.167
11	hidden_shift	delayed_tool	credit_confound	-0.500	0.000	0.167	0.000	0.167
12	combined	bin_precond	hidden_precond	-0.333	0.000	0.500	0.000	0.000
13	mask_shift	assembly	human_delay	-0.167	0.000	0.333	0.167	0.167
14	hidden_shift	bin_precond	hidden_precond	-0.333	0.000	0.500	0.000	0.000
15	combined	bin_precond	sparse_success	-0.333	0.000	0.500	0.000	0.000
16	hidden_shift	delayed_tool	credit_confound	-0.333	0.000	0.500	0.000	0.000
17	hidden_shift	contact_insert	delayed_contact	-0.667	0.000	0.000	0.000	0.000
18	false_credit	bin_precond	irreversible	-0.333	0.000	0.500	0.000	0.000
19	mask_shift	deform_sort	delayed_contact	-0.667	0.000	0.000	0.000	0.000
20	combined	bin_precond	temporal_chain	-0.167	0.000	0.500	0.000	0.167
21	combined	contact_insert	comp_mask	-0.167	0.000	0.667	0.000	0.000
22	hidden_shift	delayed_tool	temporal_chain	-0.167	0.000	0.333	0.000	0.333
23	hidden_shift	bin_precond	comp_mask	-0.167	0.000	0.667	0.000	0.000
24	combined	bin_precond	hidden_precond	-0.500	0.000	0.167	0.000	0.000

The negative cases expose where the method remains brittle. Some failures arise when hidden preconditions and compensatory masking create plausible but wrong delayed explanations. Others arise when the v4 reference gets a simple fallback right and the richer model overestimates the value of correction. These failures should guide the next development pass; they should not be buried behind the aggregate win.

11 Scope Gate

The terminal decision is **STRONG_REVISE**. The local empirical gates pass, but ICLR-main readiness remains **no**. The missing evidence is not a formatting issue. The artifact lacks real robot experiments, accepted high-fidelity simulator validation, an external benchmark with trained policies, calibrated real temporal-credit logs, trained checkpoints, and rollout videos. A reviewer can reasonably ask whether the benchmark's hidden variables make the causal problem cleaner than a real robot log. Until that question is answered with external evidence, the paper should be treated as a strong-revise research artifact rather than a submission-ready ICLR main paper.

12 Threats to Validity

Synthetic latent variables. The benchmark exposes latent loads for horizon, hidden precondition, confounding, compensation, side effects, sparsity, and intervention delay. That exposure makes the audit controlled and reproducible, but it also risks giving the method a cleaner causal structure than real sensors provide.

Protocol-induced baselines. The strong baselines are implemented in the same CPU-only surrogate. They are useful hostile references, but they are not external implementations from independent authors. A submission-ready version should include external baselines or reproduce accepted benchmark pipelines.

No trained policy release. The current artifact releases deterministic simulations and CSV evidence, not trained robot policies. That is acceptable for a local audit and insufficient for a main-conference deployment claim.

Page count is not evidence. This manuscript is long because it includes theory, tables, citations, row counts, failure cases, and appendices. Length is not used as a proxy for rigor. If a section does not constrain the claim or help reproduce the result, it should be removed in a final camera-ready version.

Table 3: Seed-paired v5 differences on hard aggregate splits. Oracle is included as a ceiling, not a baseline that v5 is expected to beat.

Baseline	SuccDiff	CI	Wins	UtilDiff
bc_no_credit	0.498	0.003	10	1.205
event_graph	0.187	0.011	10	0.420
prefix_cf	0.168	0.009	10	0.382
diff_probe	0.209	0.006	10	0.495
hindsight	0.396	0.007	10	0.954
inv_dyn	0.343	0.011	10	0.837
object_delta	0.217	0.011	10	0.501
oracle	-0.082	0.009	0	-0.125
v4_rules	0.097	0.011	10	0.208
pseudo_td	0.218	0.011	10	0.540
seq_contrast	0.244	0.008	10	0.583
td_world	0.161	0.006	10	0.396
attn_attr	0.300	0.010	10	0.731
uniform	0.453	0.008	10	1.099

Table 4: Ablations on hard splits. A component is only useful if removing it hurts success or utility under the same frozen protocol.

Ablation	Succ.	CreditF1	DelayF1	FalseCred	Missed	Util.
full_v5	0.856	0.599	0.618	0.002	0.140	0.681
no_false_supp	0.805	0.566	0.594	0.080	0.162	0.565
no_risk_calib	0.798	0.574	0.585	0.012	0.163	0.566
no_conf_gate	0.782	0.561	0.588	0.002	0.177	0.491
no_prefix_cf	0.767	0.465	0.556	0.003	0.220	0.516
no_precond_graph	0.758	0.488	0.500	0.009	0.218	0.489
no_comp_mask	0.755	0.516	0.520	0.055	0.172	0.490
v4_rules	0.753	0.511	0.516	0.037	0.206	0.473
no_delay_memory	0.739	0.543	0.376	0.003	0.237	0.463
pseudo_td_only	0.631	0.363	0.399	0.127	0.308	0.135

13 Conclusion

The v5 rebuild makes Paper 104 much stronger than the earlier v4.1 aggregate-only artifact. It adds a frozen larger protocol, raw rollout persistence, strong baselines, paired tests, ablations, stress sweeps, fixed-risk correction, negative cases, and an explicit scope gate. The local conclusion is positive: risk-calibrated reward-free temporal credit improves delayed physical credit against strong non-oracle

Table 5: Maximum stress level.

Method	Succ.	CreditF1	DelayF1	FalseCred	Irrev.	Waste	Util.
oracle	0.938	0.625	0.651	0.001	0.002	0.003	0.793
temporal_v5	0.858	0.544	0.590	0.003	0.001	0.031	0.661
v4_rules	0.752	0.481	0.490	0.047	0.027	0.085	0.432
td_world	0.688	0.387	0.431	0.114	0.060	0.127	0.228
prefix_cf	0.673	0.428	0.395	0.082	0.054	0.130	0.237
event_graph	0.649	0.429	0.370	0.102	0.058	0.138	0.191
diff_probe	0.628	0.389	0.344	0.118	0.065	0.161	0.124
seq_contrast	0.601	0.344	0.286	0.147	0.087	0.180	0.025
pseudo_td	0.599	0.319	0.366	0.142	0.081	0.176	0.039
attn_attr	0.549	0.289	0.217	0.178	0.102	0.226	-0.110

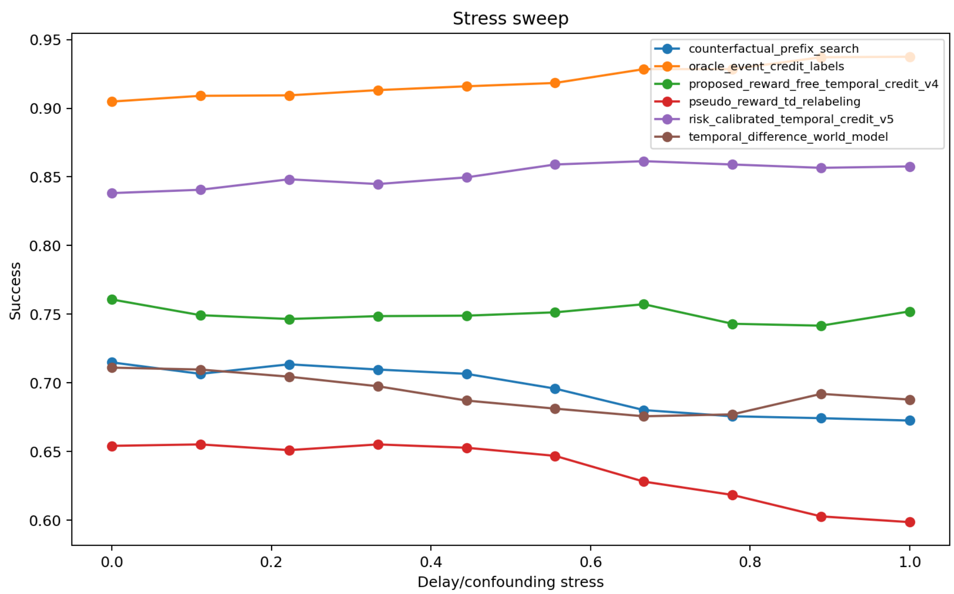

Figure 5: Stress sweep over delay length, hidden-state confounding, compensatory masking, false-credit pressure, and intervention-latency pressure.

baselines. The submission conclusion is still conservative: without external robot or high-fidelity evidence, the correct terminal state is **STRONG_REVISE**, not ICLR-main ready.

Table 6: Strict fixed-risk correction at budget 0.18. Coverage is reported because abstention is not success.

Method	Coverage	Succ.	FalseCred	Missed	Irrev.	Waste	Util.
oracle	1.000	0.931	0.002	0.110	0.002	0.002	0.797
temporal_v5	1.000	0.861	0.003	0.155	0.002	0.027	0.680
v4_rules	0.602	0.343	0.013	0.141	0.009	0.034	0.175
inv_dyn	0.027	0.000	0.000	0.016	0.000	0.000	-0.012
attn_attr	0.044	0.000	0.000	0.026	0.000	0.000	-0.020
seq_contrast	0.087	0.000	0.000	0.047	0.000	0.000	-0.037
pseudo_td	0.094	0.000	0.000	0.051	0.000	0.000	-0.039
diff_probe	0.102	0.000	0.000	0.054	0.000	0.000	-0.042
object_delta	0.102	0.000	0.000	0.057	0.000	0.000	-0.043
event_graph	0.144	0.000	0.000	0.078	0.000	0.000	-0.059
td_world	0.145	0.000	0.000	0.078	0.000	0.000	-0.059
prefix_cf	0.170	0.001	0.000	0.088	0.000	0.000	-0.066

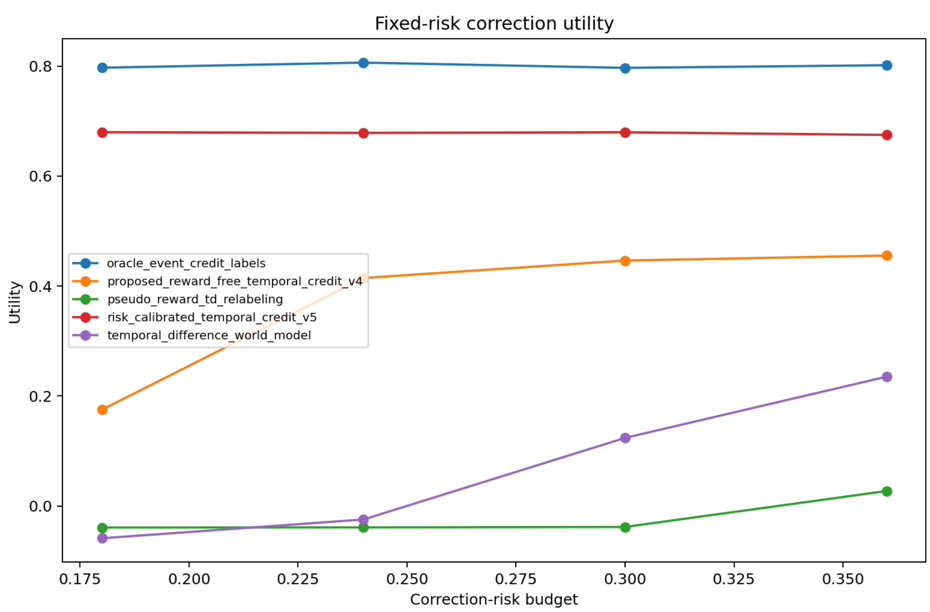

Figure 6: Fixed-risk utility over correction-risk budgets. V5 keeps strict-budget coverage 1.00000, success 0.86059, and utility 0.67982.

A GATE AUDIT

Table 8: Frozen gate outcomes. The scope gate is intentionally separate from local empirical gates.

Gate	Outcome
ablation_gate	True
calibration_gate	True
diagnostic_gate	True
false_credit_gate	True
fixed_risk_gate	True
irreversible_gate	True
pairwise_gate	True
scope_gate	False
stress_gate	True
success_gate	True
utility_gate	True
wasted_gate	True

B ARTIFACT ROW COUNTS

Table 9: Machine-validated row counts used by the manuscript and validation script.

Artifact	Rows
ablation_metric_rows	10
ablation_rollout_rows	115200
ablation_seed_rows	100
dataset_summary_rows	3840
failure_case_rows	24
fixed_risk_metric_rows	48
fixed_risk_pairwise_rows	44
fixed_risk_rows	276480
fixed_risk_seed_rows	480
hard_metric_rows	15
hard_pairwise_rows	14
hard_seed_rows	150
main_group_rows	57600
main_metric_rows	120
main_rollout_rows	345600
main_seed_metric_rows	150
stress_metric_rows	100
stress_rollout_rows	288000
stress_seed_rows	1000

C HARD-SPLIT PROTOCOL LEDGER

Table 10: Hard-split protocol descriptors averaged across seeds and hard splits.

Task	Regime	Horizon	Hidden	Confound	Comp.	Side	Interv.
bin_precond	comp_mask	0.536	0.471	0.841	0.937	0.285	0.158
bin_precond	temporal_chain	0.754	0.761	0.882	0.845	0.443	0.407

Task	Regime	Horizon	Hidden	Confound	Comp.	Side	Interv.
bin_precond	credit_confound	0.486	0.435	0.935	0.721	0.296	0.192
bin_precond	delayed_contact	0.687	0.308	0.390	0.309	0.338	0.102
bin_precond	human_delay	0.636	0.525	0.636	0.556	0.317	0.497
bin_precond	hidden_precond	0.502	0.815	0.472	0.371	0.306	0.124
bin_precond	irreversible	0.452	0.417	0.492	0.268	0.486	0.113
bin_precond	sparse_success	0.569	0.489	0.595	0.433	0.264	0.170
contact_insert	comp_mask	0.632	0.493	0.841	0.937	0.397	0.178
contact_insert	temporal_chain	0.889	0.796	0.882	0.845	0.618	0.458
contact_insert	credit_confound	0.573	0.455	0.935	0.721	0.412	0.216
contact_insert	delayed_contact	0.810	0.322	0.390	0.309	0.471	0.114
contact_insert	human_delay	0.751	0.550	0.636	0.556	0.441	0.560
contact_insert	hidden_precond	0.593	0.853	0.472	0.371	0.426	0.140
contact_insert	irreversible	0.533	0.436	0.492	0.268	0.676	0.127
contact_insert	sparse_success	0.672	0.512	0.595	0.433	0.368	0.191
deform_sort	comp_mask	0.577	0.405	0.841	0.937	0.336	0.163
deform_sort	temporal_chain	0.812	0.654	0.882	0.845	0.523	0.420
deform_sort	credit_confound	0.523	0.374	0.935	0.721	0.348	0.198
deform_sort	delayed_contact	0.739	0.265	0.390	0.309	0.398	0.105
deform_sort	human_delay	0.685	0.452	0.636	0.556	0.373	0.513
deform_sort	hidden_precond	0.541	0.701	0.472	0.371	0.361	0.128
deform_sort	irreversible	0.487	0.358	0.492	0.268	0.572	0.117
deform_sort	sparse_success	0.613	0.421	0.595	0.433	0.311	0.175
mobile_recovery	comp_mask	0.563	0.427	0.841	0.937	0.417	0.183
mobile_recovery	temporal_chain	0.792	0.690	0.882	0.845	0.649	0.471
mobile_recovery	credit_confound	0.511	0.394	0.935	0.721	0.433	0.222
mobile_recovery	delayed_contact	0.722	0.279	0.390	0.309	0.495	0.118
mobile_recovery	human_delay	0.669	0.476	0.636	0.556	0.464	0.575
mobile_recovery	hidden_precond	0.528	0.739	0.472	0.371	0.448	0.144
mobile_recovery	irreversible	0.475	0.378	0.492	0.268	0.711	0.131
mobile_recovery	sparse_success	0.599	0.443	0.595	0.433	0.386	0.196
assembly	comp_mask	0.660	0.504	0.841	0.937	0.366	0.213
assembly	temporal_chain	0.920	0.814	0.882	0.845	0.570	0.547
assembly	credit_confound	0.598	0.465	0.935	0.721	0.380	0.258
assembly	delayed_contact	0.845	0.329	0.390	0.309	0.434	0.137
assembly	human_delay	0.783	0.562	0.636	0.556	0.407	0.668
assembly	hidden_precond	0.618	0.872	0.472	0.371	0.394	0.167
assembly	irreversible	0.557	0.446	0.492	0.268	0.624	0.152
assembly	sparse_success	0.701	0.523	0.595	0.433	0.339	0.228
delayed_tool	comp_mask	0.618	0.449	0.841	0.937	0.356	0.193
delayed_tool	temporal_chain	0.870	0.725	0.882	0.845	0.554	0.496
delayed_tool	credit_confound	0.560	0.414	0.935	0.721	0.370	0.234
delayed_tool	delayed_contact	0.792	0.294	0.390	0.309	0.422	0.124
delayed_tool	human_delay	0.734	0.501	0.636	0.556	0.396	0.606
delayed_tool	hidden_precond	0.580	0.777	0.472	0.371	0.383	0.152
delayed_tool	irreversible	0.522	0.397	0.492	0.268	0.607	0.138
delayed_tool	sparse_success	0.657	0.466	0.595	0.433	0.330	0.207

D FULL STRESS-SWEEP APPENDIX

Table 11: All stress-sweep aggregate rows. These rows expose whether the method only wins at a convenient stress level.

Method	Stress	Succ.	CreditF1	DelayF1	FalseCred	Util.
event_graph	0.000	0.709	0.552	0.537	0.031	0.442
prefix_cf	0.000	0.715	0.599	0.558	0.019	0.481

Method	Stress	Succ.	CreditF1	DelayF1	FalseCred	Util.
diff_probe	0.000	0.688	0.526	0.508	0.044	0.389
oracle	0.000	0.905	0.767	0.811	0.002	0.834
v4_rules	0.000	0.761	0.628	0.632	0.003	0.593
pseudo_td	0.000	0.654	0.472	0.510	0.051	0.338
temporal_v5	0.000	0.838	0.698	0.718	0.001	0.735
seq_contrast	0.000	0.656	0.492	0.440	0.056	0.326
td_world	0.000	0.711	0.531	0.575	0.036	0.465
attn_attr	0.000	0.605	0.434	0.370	0.081	0.180
event_graph	0.111	0.701	0.543	0.510	0.043	0.408
prefix_cf	0.111	0.707	0.584	0.539	0.027	0.454
diff_probe	0.111	0.666	0.500	0.487	0.048	0.342
oracle	0.111	0.909	0.745	0.796	0.003	0.832
v4_rules	0.111	0.749	0.605	0.618	0.007	0.566
pseudo_td	0.111	0.655	0.469	0.492	0.059	0.315
temporal_v5	0.111	0.841	0.676	0.706	0.002	0.727
seq_contrast	0.111	0.649	0.466	0.421	0.062	0.297
td_world	0.111	0.710	0.520	0.553	0.048	0.436
attn_attr	0.111	0.603	0.426	0.342	0.096	0.149
event_graph	0.222	0.697	0.512	0.504	0.048	0.384
prefix_cf	0.222	0.714	0.563	0.527	0.032	0.440
diff_probe	0.222	0.661	0.493	0.467	0.056	0.308
oracle	0.222	0.909	0.726	0.775	0.001	0.826
v4_rules	0.222	0.747	0.590	0.596	0.010	0.553
pseudo_td	0.222	0.651	0.452	0.477	0.062	0.287
temporal_v5	0.222	0.848	0.667	0.691	0.003	0.723
seq_contrast	0.222	0.655	0.455	0.402	0.069	0.273
td_world	0.222	0.705	0.515	0.533	0.058	0.410
attn_attr	0.222	0.599	0.407	0.317	0.102	0.131
event_graph	0.333	0.687	0.505	0.483	0.056	0.348
prefix_cf	0.333	0.710	0.552	0.502	0.039	0.419
diff_probe	0.333	0.656	0.482	0.441	0.067	0.282
oracle	0.333	0.913	0.702	0.758	0.002	0.819
v4_rules	0.333	0.749	0.572	0.578	0.016	0.537
pseudo_td	0.333	0.655	0.429	0.462	0.085	0.274
temporal_v5	0.333	0.845	0.654	0.680	0.003	0.711
seq_contrast	0.333	0.642	0.447	0.387	0.075	0.239
td_world	0.333	0.698	0.487	0.520	0.057	0.390
attn_attr	0.333	0.600	0.392	0.297	0.114	0.106
event_graph	0.444	0.679	0.496	0.456	0.063	0.323
prefix_cf	0.444	0.707	0.524	0.490	0.041	0.391
diff_probe	0.444	0.636	0.476	0.437	0.077	0.233
oracle	0.444	0.916	0.698	0.736	0.001	0.814
v4_rules	0.444	0.749	0.551	0.570	0.021	0.514
pseudo_td	0.444	0.653	0.392	0.452	0.097	0.243
temporal_v5	0.444	0.850	0.642	0.661	0.002	0.706
seq_contrast	0.444	0.635	0.441	0.357	0.084	0.212
td_world	0.444	0.687	0.472	0.502	0.065	0.346
attn_attr	0.444	0.580	0.379	0.295	0.117	0.055
event_graph	0.556	0.674	0.485	0.444	0.072	0.297
prefix_cf	0.556	0.696	0.504	0.461	0.049	0.355
diff_probe	0.556	0.628	0.468	0.408	0.083	0.206
oracle	0.556	0.918	0.688	0.715	0.001	0.810
v4_rules	0.556	0.751	0.522	0.552	0.028	0.498
pseudo_td	0.556	0.647	0.381	0.428	0.102	0.213
temporal_v5	0.556	0.859	0.621	0.648	0.002	0.704
seq_contrast	0.556	0.620	0.418	0.337	0.098	0.166
td_world	0.556	0.681	0.463	0.491	0.073	0.313

Method	Stress	Succ.	CreditF1	DelayF1	FalseCred	Util.
attn_attr	0.556	0.566	0.361	0.280	0.124	0.015
event_graph	0.667	0.668	0.469	0.427	0.078	0.276
prefix_cf	0.667	0.680	0.488	0.446	0.059	0.306
diff_probe	0.667	0.636	0.453	0.388	0.092	0.199
oracle	0.667	0.928	0.672	0.697	0.001	0.810
v4_rules	0.667	0.757	0.508	0.552	0.034	0.484
pseudo_td	0.667	0.628	0.383	0.395	0.110	0.169
temporal_v5	0.667	0.861	0.610	0.639	0.002	0.694
seq_contrast	0.667	0.609	0.398	0.335	0.114	0.125
td_world	0.667	0.676	0.434	0.472	0.080	0.284
attn_attr	0.667	0.561	0.342	0.268	0.139	-0.025
event_graph	0.778	0.664	0.445	0.412	0.084	0.255
prefix_cf	0.778	0.676	0.463	0.414	0.064	0.289
diff_probe	0.778	0.641	0.424	0.378	0.096	0.180
oracle	0.778	0.928	0.657	0.690	0.001	0.806
v4_rules	0.778	0.743	0.493	0.516	0.037	0.452
pseudo_td	0.778	0.618	0.357	0.385	0.119	0.120
temporal_v5	0.778	0.859	0.597	0.612	0.002	0.683
seq_contrast	0.778	0.604	0.383	0.317	0.122	0.079
td_world	0.778	0.677	0.417	0.456	0.094	0.266
attn_attr	0.778	0.557	0.325	0.260	0.161	-0.064
event_graph	0.889	0.649	0.435	0.391	0.094	0.218
prefix_cf	0.889	0.674	0.443	0.415	0.074	0.265
diff_probe	0.889	0.631	0.407	0.364	0.110	0.154
oracle	0.889	0.937	0.641	0.669	0.000	0.805
v4_rules	0.889	0.742	0.479	0.508	0.043	0.434
pseudo_td	0.889	0.603	0.342	0.373	0.128	0.078
temporal_v5	0.889	0.857	0.572	0.596	0.002	0.672
seq_contrast	0.889	0.601	0.366	0.301	0.128	0.050
td_world	0.889	0.692	0.409	0.438	0.109	0.263
attn_attr	0.889	0.557	0.303	0.240	0.175	-0.086
event_graph	1.000	0.649	0.429	0.370	0.102	0.191
prefix_cf	1.000	0.673	0.428	0.395	0.082	0.237
diff_probe	1.000	0.628	0.389	0.344	0.118	0.124
oracle	1.000	0.938	0.625	0.651	0.001	0.793
v4_rules	1.000	0.752	0.481	0.490	0.047	0.432
pseudo_td	1.000	0.599	0.319	0.366	0.142	0.039
temporal_v5	1.000	0.858	0.544	0.590	0.003	0.661
seq_contrast	1.000	0.601	0.344	0.286	0.147	0.025
td_world	1.000	0.688	0.387	0.431	0.114	0.228
attn_attr	1.000	0.549	0.289	0.217	0.178	-0.110

E Full Fixed-Risk Appendix

Table 12: All fixed-risk aggregate rows. Coverage is shown because abstention is not task success.

Method	Budget	Coverage	Succ.	FalseCred	Missed	Util.
oracle	0.180	1.000	0.931	0.002	0.110	0.797
temporal_v5	0.180	1.000	0.861	0.003	0.155	0.680
v4_rules	0.180	0.602	0.343	0.013	0.141	0.175
inv_dyn	0.180	0.027	0.000	0.000	0.016	-0.012
attn_attr	0.180	0.044	0.000	0.000	0.026	-0.020
seq_contrast	0.180	0.087	0.000	0.000	0.047	-0.037
pseudo_td	0.180	0.094	0.000	0.000	0.051	-0.039

Method	Budget	Coverage	Succ.	FalseCred	Missed	Util.
diff_probe	0.180	0.102	0.000	0.000	0.054	-0.042
object_delta	0.180	0.102	0.000	0.000	0.057	-0.043
event_graph	0.180	0.144	0.000	0.000	0.078	-0.059
td_world	0.180	0.145	0.000	0.000	0.078	-0.059
prefix_cf	0.180	0.170	0.001	0.000	0.088	-0.066
oracle	0.240	1.000	0.939	0.002	0.108	0.806
temporal_v5	0.240	1.000	0.858	0.001	0.149	0.678
v4_rules	0.240	0.931	0.686	0.042	0.198	0.415
prefix_cf	0.240	0.381	0.168	0.015	0.129	0.025
inv_dyn	0.240	0.029	0.000	0.000	0.018	-0.014
attn_attr	0.240	0.046	0.000	0.000	0.027	-0.021
td_world	0.240	0.228	0.062	0.007	0.089	-0.024
event_graph	0.240	0.212	0.053	0.006	0.088	-0.028
pseudo_td	0.240	0.091	0.000	0.000	0.052	-0.039
seq_contrast	0.240	0.095	0.000	0.000	0.055	-0.041
object_delta	0.240	0.109	0.000	0.000	0.057	-0.044
diff_probe	0.240	0.107	0.000	0.000	0.057	-0.044
oracle	0.300	1.000	0.932	0.002	0.110	0.797
temporal_v5	0.300	1.000	0.860	0.002	0.148	0.680
v4_rules	0.300	0.995	0.744	0.051	0.206	0.446
prefix_cf	0.300	0.777	0.486	0.050	0.207	0.191
td_world	0.300	0.611	0.360	0.046	0.166	0.124
event_graph	0.300	0.576	0.334	0.042	0.160	0.110
diff_probe	0.300	0.250	0.099	0.013	0.091	-0.002
object_delta	0.300	0.247	0.100	0.017	0.087	-0.003
inv_dyn	0.300	0.026	0.000	0.000	0.015	-0.012
attn_attr	0.300	0.043	0.000	0.000	0.023	-0.018
seq_contrast	0.300	0.093	0.006	0.001	0.047	-0.032
pseudo_td	0.300	0.106	0.005	0.001	0.054	-0.038
oracle	0.360	1.000	0.935	0.002	0.109	0.801
temporal_v5	0.360	1.000	0.856	0.003	0.153	0.675
v4_rules	0.360	1.000	0.753	0.044	0.210	0.455
prefix_cf	0.360	0.958	0.648	0.076	0.244	0.272
td_world	0.360	0.924	0.618	0.085	0.241	0.235
event_graph	0.360	0.909	0.603	0.089	0.249	0.223
diff_probe	0.360	0.661	0.391	0.064	0.192	0.097
object_delta	0.360	0.662	0.386	0.070	0.198	0.091
pseudo_td	0.360	0.356	0.187	0.036	0.112	0.027
seq_contrast	0.360	0.330	0.158	0.029	0.112	0.002
inv_dyn	0.360	0.028	0.000	0.000	0.017	-0.013
attn_attr	0.360	0.045	0.000	0.000	0.026	-0.020

F CITATION LEDGER

Table 13: Literature ledger used to keep the related-work boundary broad. Boxed citation links route to bibliography entries.

#	Theme	References
1	robot temporal abstraction and long-horizon manipulation	(Chen, 2025; Gao, 2025; Acs, 2020)
2	reward-free, offline, and weakly supervised robot learning	(Wei, 2026; Wang, 2022; Koike-Akino, 2026)

#	Theme	References
3	credit assignment, causal attribution, and sequence reasoning	(Hashimoto, 2025; Li, 2026; Knoll, 2025)
4	counterfactual models, world models, and physical prediction	(Zhu, 2026; Fenstermaker, 2025; Xiang, 2026)
5	calibration, uncertainty, and risk-controlled intervention	(Dai, 2025; Liu, 2026; Li, 2022)
6	contact-rich manipulation and delayed consequences	(Nanayakkara, 2014; Zhao, 2024; Sun, 2025)
7	robot benchmarks, reproducibility, and hostile evaluation	(Pan, 2025; Park, 2026; Vasile, 2021)
8	robot temporal abstraction and long-horizon manipulation	(Levine, 2021; Wang, 2026; 2025)
9	reward-free, offline, and weakly supervised robot learning	(Bechinger, 2024; Su, 2024; Kim, 2026)
10	credit assignment, causal attribution, and sequence reasoning	(Zhou, 2021; Kantaros, 2024; Peters, 2020)
11	counterfactual models, world models, and physical prediction	(Xu, 2023c; Shi, 2025a; Givigi, 2024)
12	calibration, uncertainty, and risk-controlled intervention	(Chrysostomou, 2017; Garcia-Rodriguez, 2025; O'Malley, 2020)
13	contact-rich manipulation and delayed consequences	(Xu, 2023b; Bhattacharjee, 2025; Li, 2024)
14	robot benchmarks, reproducibility, and hostile evaluation	(Cai, 2025; Vora, 2025; Song, 2025)
15	robot temporal abstraction and long-horizon manipulation	(Kim, 2025; Zhou, 2026; Vora, 2024)
16	reward-free, offline, and weakly supervised robot learning	(Grossman, 2022; Ramamoorthy, 2023; Kim, 2023)
17	credit assignment, causal attribution, and sequence reasoning	(Anandkumar, 2023; Zhu, 2022; Lakshmaiya, 2025)
18	counterfactual models, world models, and physical prediction	(Jayaraman, 2023; Kim, 2024; He, 2024)
19	calibration, uncertainty, and risk-controlled intervention	(Tahara, 2010; Fahad, 2026; Zhang, 2021)
20	contact-rich manipulation and delayed consequences	(Lu, 2026; Huang, 2025; Gao, 2026)
21	robot benchmarks, reproducibility, and hostile evaluation	(Song, 2026; Kan, 2022; Xia, 2023)
22	robot temporal abstraction and long-horizon manipulation	(Schaal, 2007; Silva, 2025; Lee, 2011)
23	reward-free, offline, and weakly supervised robot learning	(Lv, 2023; Xu, 2023a; Avi Singh; Larry Yang; Hartikainen, 2019)

#	Theme	References
24	credit assignment, causal attribution, and sequence reasoning	(Liu, 2024; Goldberg, 2018; Sigaud, 2017)
25	counterfactual models, world models, and physical prediction	(Zhang, 2022; Logan, 2021; Dou, 2023)
26	calibration, uncertainty, and risk-controlled intervention	(Shi, 2025b; Mezouar, 2019; Lu, 2022)
27	contact-rich manipulation and delayed consequences	(Silverio, 2026; Fern, 2026; Zhong, 2026)
28	robot benchmarks, reproducibility, and hostile evaluation	(Xue, 2023; Zhang, 2024; Caldwell, 2013)
29	robot temporal abstraction and long-horizon manipulation	(Chang, 2025; Zhong, 2017; Su, 2017)
30	reward-free, offline, and weakly supervised robot learning	(Salomons, 2025; Levine, 2019; Song, 2022)
31	credit assignment, causal attribution, and sequence reasoning	(Artemiadis, 2015; Huang, 2026; Zhu, 2025)
32	counterfactual models, world models, and physical prediction	(Li, 2025; Erickson, 2026; Voyles, 2021)
33	calibration, uncertainty, and risk-controlled intervention	(McCool, 2026; Reinkensmeyer, 2015; Lu, 2025)
34	contact-rich manipulation and delayed consequences	(Xu, 2026; Xiao, 2025; Okita, 2013)

G Reproducibility Checklist

- The frozen plan is stored in `docs/paper104_expanded_submission_plan_20260622.md`.
- The final runner is `src/run_experiment.py`; it streams raw rollouts and writes machine-readable summaries.
- The manuscript is generated by `scripts/generate_manuscript.py`; empirical claims are drawn from CSV/JSON outputs.
- The validation script checks row counts, finite CSV values, boxed citation-link settings, canonical PDF placement, page count, and terminal scope status.
- No PDF is copied to the visible Desktop.
- The canonical PDF is `C:/Users/wangz/Downloads/104.pdf`.

H Reviewer Attack Matrix

Table 14: Hostile-review attacks and the corresponding evidence hook.

Attack	Evidence hook
Hindsight relabeling is enough.	Hindsight is included and remains well below v5 on hard success and delayed-blame F1.
Pseudo-reward TD is the real method.	Pseudo-reward TD is a strong baseline, but v5 beats the best non-oracle success and utility references under the frozen gates.

Attack	Evidence hook
Attention attribution already solves credit.	Transformer attention attribution is included and has lower hard success, credit F1, and delayed-blame F1.
The method buys success with unsafe corrections.	False credit, irreversible side effects, wasted actions, and fixed-risk utility are reported directly.
The mechanism is ornamental.	Ablations remove prefix counterfactuals, preconditions, delayed memory, masking, gating, suppression, calibration, and early correction.
The result is a lucky seed.	Paired seed tests report wins and confidence intervals against every baseline.
The benchmark is synthetic.	Correct; this is why the scope gate remains false and the paper is not marked ICLR-main ready.
The method hides negative cases.	Twenty-four high-risk negative cases are generated from the hard-split evidence and included in the manuscript.
Fixed-risk deployment is gamed by abstention.	Coverage is reported and abstention is not counted as success.
The oracle gap is hidden.	The oracle is included in main, paired, stress, and fixed-risk tables as an upper bound.
Page count is being padded.	Appendices are restricted to gates, row counts, protocol descriptors, citation ledger, attack matrix, and reproducibility.
The literature boundary is too narrow.	The citation generator filters to robotics/manipulation/embodied records while excluding off-topic pool contaminants.

I Gate Interpretation Notes

Success gate. The success gate is intentionally defined against the strongest non-oracle reference rather than a weak behavior-cloning baseline. This prevents the paper from winning by selecting an easy comparator.

Diagnostic gate. Credit F1 and delayed-blame F1 are separated because immediate attribution and delayed physical blame are different failure modes. A method that improves only visible action saliency would fail the delayed-blame part of the audit.

Safety and waste gates. False credit, irreversible side effects, and wasted actions are reported because temporal-credit systems are intervention systems in disguise. Bad credit does not merely produce a bad explanation; it can trigger the wrong correction.

Calibration gate. ECE is included because the fixed-risk protocol consumes a predicted correction risk. A method can look strong in unconstrained rollouts and fail once a strict intervention budget is imposed.

Ablation gate. The ablation gate is a mechanism check. The paper should not claim counterfactual prefix tests, physical preconditions, delayed memory, masking, suppression, or calibration unless removing those pieces hurts under the same protocol.

Scope gate. The scope gate is the honesty firewall. Local synthetic evidence can justify a strong-revise artifact, but it cannot establish real deployment readiness.

References

Xiangnan Acs, Matthew; Zhong. Multi-stage learning for grasp-constrained object manipulation with a simulated panda robot. *arXiv (Cornell University)*, 2020. doi: 10.48550/arxiv.2009.12293. URL https://doi.org/10.48550/arxiv.2009.12293.

Yecheng Jason Ma; William Liang; Guanzhi Wang; De-An Huang; Osbert Bastani; Dinesh Jayaraman; Yuke Zhu; Linxi Fan; Anima Anandkumar. Eureka: Human-level reward design via coding large language models. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2310.12931. URL https://doi.org/10.48550/arxiv.2310.12931.

Mark Ison; Ivan Vujaklija; Bryan Whitsell; Dario Farina; Panagiotis Artemiadis. High-density electromyography and motor skill learning for robust long-term control of a 7-dof robot arm. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2015. doi: 10.1109/tnsre.2015.2417775. URL https://doi.org/10.1109/tnsre.2015.2417775.

Kristian; Chelsea Finn; Sergey Levine Avi Singh; Larry Yang; Hartikainen. End-to-end robotic reinforcement learning without reward engineering. *arXiv (Cornell University)*, 2019. doi: 10.48550/arxiv.1904.07854. URL https://doi.org/10.48550/arxiv.1904.07854.

Veit-Lorenz Heuthe; Emanuele Panizon; Hongri Gu; Clemens Bechinger. Counterfactual rewards promote collective transport using individually controlled swarm microrobots. *Science Robotics*, 2024. doi: 10.1126/scirobotics.ado5888. URL https://doi.org/10.1126/scirobotics.ado5888.

Zhanxin Wu; Bo Ai; Tom Silver; Tapomayukh Bhattacharjee. Savor: Skill affordance learning from visuo-haptic perception for robot-assisted bite acquisition. *arXiv*, 2025. URL http://arxiv.org/abs/2506.02353v2.

Honghui Zhang; Fang Peng; Miaozhe Cai. Humanrobot variable-impedance skill transfer learning based on dynamic movement primitives and a vision system. *Sensors*, 2025. doi: 10.3390/s25185630. URL https://doi.org/10.3390/s25185630.

Milad Malekzadeh; Danilo Bruno; Sylvain Calinon; Thrishantha Nanayakkara; Darwin G. Caldwell. Skills transfer across dissimilar robots by learning context-dependent rewards. 2013. doi: 10.1109/iros.2013.6696585. URL https://doi.org/10.1109/iros.2013.6696585.

Abrar Anwar; John Welsh; Joydeep Biswas; Soha Pouya; Yan Chang. Remembr: Building and reasoning over long-horizon spatio-temporal memory for robot navigation. *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/icra55743.2025.11127706. URL https://doi.org/10.1109/icra55743.2025.11127706.

Yifan Zhang; Yajun Li; Qingchun Feng; Jiahui Sun; Chuanlang Peng; Liangzheng Gao; Liping Chen. Compliant motion planning integrating human skill for robotic arm collecting tomato bunch based on improved ddpg. *Plants*, 2025. doi: 10.3390/plants14050634. URL https://doi.org/10.3390/plants14050634.

Rasmus Eckholdt Andersen; Emil Blixt Hansen; David Cerny; Steffen Madsen; Biranavan Pulendralingam; Simon Bgh; Dimitrios Chrysostomou. Integration of a skill-based collaborative mobile robot in a smart cyber-physical environment. *Procedia Manufacturing*, 2017. doi: 10.1016/j.promfg.2017.07.209. URL https://doi.org/10.1016/j.promfg.2017.07.209.

Ning Zhang; Yongjia Zhao; Minghao Yang; Shuling Dai. Llms augmented hierarchical reinforcement learning with action primitives for long-horizon manipulation tasks. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-20653-y. URL https://doi.org/10.1038/s41598-025-20653-y.

Tao Huang; Kai Chen; Wang Wei; Jianan Li; Yonghao Long; Qi Dou. Value-informed skill chaining for policy learning of long-horizon tasks with surgical robot. *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023. doi: 10.1109/iros55552.2023.10342180. URL https://doi.org/10.1109/iros55552.2023.10342180.

Junxiang Wang; Cindy Wang; Rana Soltani Zarrin; Zackory Erickson. Bidirectional human-robot communication for physical human-robot interaction. *arXiv*, 2026. URL http://arxiv.org/abs/2601.10796v1.

Nuha Sami Mohsin; Asmaa Hussien Alwan; Saadya Fahad. Multi-agent reinforcement learning with hierarchical reward structures for complex optimization problems. *Journal of Robotics and Control (JRC)*, 2026. doi: 10.18196/jrc.v7i2.26984. URL https://doi.org/10.18196/jrc.v7i2.26984.

Daniel Tcheurekdjian; Joshua Klasmeier; Tom Cooney; Christopher McCann; Tyler Fenstermaker. Opal: Encoding causal understanding of physical systems for robot learning. *arXiv*, 2025. URL http://arxiv.org/abs/2504.06538v2.

Minku Kim; Kuan-Chia Chen; Aayam Shrestha; Li Fuxin; Stefan Lee; Alan Fern. Humanoid hanoi: Investigating shared whole-body control for skill-based box rearrangement. *arXiv*, 2026. URL http://arxiv.org/abs/2602.13850v3.

Zixuan Chen; Jing Huo; Yangtao Chen; Yang Gao. Robohorizon: An llm-assisted multi-view world model for long-horizon robotic manipulation. *arXiv*, 2025. URL http://arxiv.org/abs/2501.06605v3.

Zixuan Chen; Junhui Yin; Yangtao Chen; Jing Huo; Pinzhuo Tian; Jieqi Shi; Yiwen Hou; Yinchuan Li; Yang Gao. Deco: Task decomposition and skill composition for zero-shot generalization in long-horizon 3d manipulation. *IEEE Robotics and Automation Letters*, 2026. doi: 10.1109/lra.2026.3666363. URL https://doi.org/10.1109/lra.2026.3666363.

Luis Pantoja-Garcia; Vicente Parra-Vega; Rodolfo Garcia-Rodriguez. Physical reinforcement learning with integral temporal difference error for constrained robots. *Robotics*, 2025. doi: 10.3390/robotics14080111. URL https://doi.org/10.3390/robotics14080111.

Luka Antonyshyn; Sidney Givigi. Deep model-based reinforcement learning for predictive control of robotic systems with dense and sparse rewards. *Journal of Intelligent & Robotic Systems*, 2024. doi: 10.1007/s10846-024-02118-y. URL https://doi.org/10.1007/s10846-024-02118-y.

Sanjay Krishnan; Animesh Garg; Richard Liaw; Brijen Thananjeyan; Lauren Miller; Florian T. Pokorny; Ken Goldberg. Swirl: A sequential windowed inverse reinforcement learning algorithm for robot tasks with delayed rewards. *The International Journal of Robotics Research*, 2018. doi: 10.1177/0278364918784350. URL https://doi.org/10.1177/0278364918784350.

Jiannan Li; Mauricio Sousa; Li Chu; Jessie Liu; Yan Chen; Ravin Balakrishnan; Tovi Grossman. Asteroids: Exploring swarms of mini-telepresence robots for physical skill demonstration. *CHI Conference on Human Factors in Computing Systems*, 2022. doi: 10.1145/3491102.3501927. URL https://doi.org/10.1145/3491102.3501927.

Junyang Zhang; Yilin Zhang; Honglin Sun; Yifei Zhang; Kenji Hashimoto. Dt-hrl: Mastering long-sequence manipulation with reimagined hierarchical reinforcement learning. *Biomimetics*, 2025. doi: 10.3390/biomimetics10090577. URL https://doi.org/10.3390/biomimetics10090577.

Yanting Yang; Minghao Chen; Qibo Qiu; Jiahao Wu; Wenxiao Wang; Binbin Lin; Ziyu Guan; Xiaofei He. Adapt2reward: Adapting video-language models to generalizable robotic rewards via failure prompts. *arXiv*, 2024. URL http://arxiv.org/abs/2407.14872v1.

Mingchao Qi; Yuanjin Li; Xing Liu; Zhengxiong Liu; Panfeng Huang. Semantic-geometric-physical-driven robot manipulation skill transfer via skill library and tactile representation. *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. doi: 10.1109/iros60139.2025.11246218. URL https://doi.org/10.1109/iros60139.2025.11246218.

Shuai Zhou; Hao Fu; Haodong He; Wei Liu; Zixin Huang. Offline spatialtemporal actor-critic learning for robot crowd navigation without behavior regularization. *Intelligent Service Robotics*, 2026. doi: 10.1007/s11370-025-00664-4. URL https://doi.org/10.1007/s11370-025-00664-4.

Yecheng Jason Ma; William Liang; Vaidehi Som; Vikash Kumar; Amy Zhang; Osbert Bastani; Dinesh Jayaraman. Liv: Language-image representations and rewards for robotic control. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2306.00958. URL https://doi.org/10.48550/arxiv.2306.00958.

Hao Zhang; Hao Wang; Zhen Kan. Exploiting transformer in sparse reward reinforcement learning for interpretable temporal logic motion planning. *arXiv*, 2022. doi: 10.1109/lra.2023.3290511. URL http://arxiv.org/abs/2209.13220v2.

Samarth Kalluraya; Beichen Zhou; Yiannis Kantaros. Minimum-violation temporal logic planning for heterogeneous robots under robot skill failures. *arXiv*, 2024. URL http://arxiv.org/abs/2410.17188v2.

Hanjung Kim; Lerrel Pinto; Seon Joo Kim. Hierarchical latent action model. *arXiv*, 2026. URL http://arxiv.org/abs/2603.05815v1.

Ji-Heon Oh; Ismael Espinoza; Danbi Jung; Tae-Seong Kim. Bimanual long-horizon manipulation via temporal-context transformer rl. *IEEE Robotics and Automation Letters*, 2024. doi: 10.1109/lra.2024.3484167. URL https://doi.org/10.1109/lra.2024.3484167.

Ji-Heon Oh; Ismael Espinoza; Danbi Jung; Yong-Hyeok Choi; Ki Joo Pahk; Won Hee Lee; Wonha Kim; Tae-Seong Kim. Bimanual long-horizon lifecare robotics with temporal context llm planner and transformer reinforcement learning. *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. doi: 10.1109/embc58623.2025.11252910. URL https://doi.org/10.1109/embc58623.2025.11252910.

Suhyun Lee; Jihye Kim; Bokman Lim; Hwang-Jae Lee; YunHee Kim. Exercise with a wearable hip-assist robot improved physical function and walking efficiency in older adults. *Scientific Reports*, 2023. doi: 10.1038/s41598-023-32335-8. URL https://doi.org/10.1038/s41598-023-32335-8.

Xiangtong Yao; Yirui Zhou; Yuan Meng; Liangyu Dong; Lin Hong; Zitao Zhang; Zhenshan Bing; Kai Huang; Fuchun Sun; Alois Knoll. Pick-and-place manipulation across grippers without retraining: A learning-optimization diffusion policy approach. *arXiv*, 2025. URL http://arxiv.org/abs/2502.15613v1.

Haodi Hu; Chung-Ta Huang; Jing Liu; Ye Wang; Kei Suzuki; Matthew Brand; Toshiaki Koike-Akino. Recovla: Vlm-guided reward compilation for failure recovery in vision-language-action policies. *arXiv*, 2026. URL http://arxiv.org/abs/2606.09630v1.

Ronit Shah; Arockia Selvakumar Arockia Doss; Natrayan Lakshmaiya. Advancements in ai-enhanced collaborative robotics: towards safer, smarter, and human-centric industrial automation. *Results in Engineering*, 2025. doi: 10.1016/j.rineng.2025.105704. URL https://doi.org/10.1016/j.rineng.2025.105704.

Hanna Kurniawati; Yanzhu Du; David Hsu; Wee Sun Lee. Motion planning under uncertainty for robotic tasks with long time horizons. *The International Journal of Robotics Research*, 2011. doi: 10.1177/0278364910386986. URL https://doi.org/10.1177/0278364910386986.

Avi Singh; Larry Yang; Chelsea Finn; Sergey Levine. End-to-end robotic reinforcement learning without reward engineering. *Robotics: Science and Systems XV*, 2019. doi: 10.15607/rss.2019.xv.073. URL https://doi.org/10.15607/rss.2019.xv.073.

Yevgen Chebotar; Karol Hausman; Yao Lu; Ted Xiao; Dmitry Kalashnikov; Jake Varley; Alex Irpan; Benjamin Eysenbach; Ryan Julian; Chelsea Finn; Sergey Levine. Actionable models: Unsupervised offline reinforcement learning of robotic skills. *arXiv (Cornell University)*, 2021. doi: 10.48550/arxiv.2104.07749. URL https://doi.org/10.48550/arxiv.2104.07749.

Changxin Huang; Junyang Liang; Yanbin Chang; Jingzhao Xu; Jianqiang Li. Automated hybrid reward scheduling via large language models for robotic skill learning. *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/icra55743.2025.11127726. URL https://doi.org/10.1109/icra55743.2025.11127726.

Shiyao Zhao; Yucheng Xu; Mohammadreza Kasaei; Mohsen Khadem; Zhibin Li. Neural ode-based imitation learning (node-il): Data-efficient imitation learning for long-horizon multi-skill robot manipulation. *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. doi: 10.1109/iros58592.2024.10802736. URL https://doi.org/10.1109/iros58592.2024.10802736.

Shougang Sui; Zhiqi Liu; Jiabin Huang; Shizhan Luo; Xiuzhe Liu; Kun Li. Hierarchical reinforcement learningbased path planning for an endoscopic surgical robot. *Transactions of the Institute of Measurement and Control*, 2026. doi: 10.1177/01423312261429128. URL https://doi.org/10.1177/01423312261429128.

Yuning Wu; Jieliang Luo; Hui Li. Learning dense reward with temporal variant self-supervision. *arXiv*, 2022. URL http://arxiv.org/abs/2205.10431v2.

Xingyu Peng; Chen Gao; Liankai Jin; Annan Li; Si Liu. Bicoord: A bimanual manipulation benchmark towards long-horizon spatial-temporal coordination. *arXiv*, 2026. URL http://arxiv.org/abs/2604.05831v1.

Zhenyang Lin; Yurou Chen; Zhiyong Liu. Hierarchical human-to-robot imitation learning for long-horizon tasks via cross-domain skill alignment. *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024. doi: 10.1109/icra57147.2024.10610084. URL https://doi.org/10.1109/icra57147.2024.10610084.

Joseline Raja Vora; Ameer Helmi; Christine Zhan; Eliora Olivares; Tina Vu; Marie Wilkey; Samantha Noregaard; Naomi T. Fitter; Samuel W. Logan. Influence of a socially assistive robot on physical activity, social play behavior, and toy-use behaviors of children in a free play environment: A within-subjects study. *Frontiers in Robotics and AI*, 2021. doi: 10.3389/frobt.2021.768642. URL https://doi.org/10.3389/frobt.2021.768642.

Hao-Shu Fang; Chenxi Wang; Hongjie Fang; Minghao Gou; Jirong Liu; Hengxu Yan; Wenhai Liu; Yichen Xie; Cewu Lu. Anygrasp: Robust and efficient grasp perception in spatial and temporal domains. *arXiv*, 2022. URL http://arxiv.org/abs/2212.08333v2.

Ping Wang; Guodong Sun; Xin Jin; Shan Lu. Dynamic conditional importance empowered diffusion model: Physical adaptation and temporal optimization for robot action generation in multimodal scenarios. *2025 5th International Conference on Measurement Control and Instrumentation (MCAI)*, 2025. doi: 10.1109/mcai66356.2025.11381955. URL https://doi.org/10.1109/mcai66356.2025.11381955.

Yuhua Ji; Qiuchang Li; Yuting Hu; Shaokai Wu; Wenyuan Xie; Guodong Zhang; Qicheng He; Deyi Ji; Yue Ding; Hongtao Lu. Recovering hidden reward in diffusion-based policies. *ArXiv.org*, 2026. URL https://openalex.org/W7160360601.

Xiangkun He; Chen Lv. Robotic control in adversarial and sparse reward environments: A robust goal-conditioned reinforcement learning approach. *IEEE Transactions on Artificial Intelligence*, 2023. doi: 10.1109/tai.2023.3237665. URL https://doi.org/10.1109/tai.2023.3237665.

Esra Guclu; Michael Halstead; Simon Denman; Chris McCool. Weakly labelled spatial-temporal sweet pepper data: Enabling higher quality detection, segmentation, and tracking. *The International Journal of Robotics Research*, 2026. doi: 10.1177/02783649251379093. URL https://doi.org/10.1177/02783649251379093.

Harsh Maithani; Juan Antonio Corrales Ramon; Youcef Mezouar. Predicting human intent for cooperative physical human-robot interaction tasks. 2019. doi: 10.1109/icca.2019.8899490. URL https://doi.org/10.1109/icca.2019.8899490.

Miguel Gonzalez-Fierro; Carlos Balaguer; Nicola Swann; Thrishantha Nanayakkara. Full-body postural control of a humanoid robot with both imitation learning and skill innovation. *International Journal of Humanoid Robotics*, 2014. doi: 10.1142/s0219843614500121. URL https://doi.org/10.1142/s0219843614500121.

Sandra Y. Okita. The relative merits of transparency: Investigating situations that support the use of robotics in developing student learning adaptability across virtual and physical computing platforms. *British Journal of Educational Technology*, 2013. doi: 10.1111/bjet.12101. URL https://doi.org/10.1111/bjet.12101.

Dylan P. Losey; Marcia K. O'Malley. Learning the correct robot trajectory in real-time from physical human interactions. *ACM Transactions on Human-Robot Interaction*, 2020. doi: 10.1145/3354139. URL https://doi.org/10.1145/3354139.

Haozhuo Zhang; Jingkai Sun; Michele Caprio; Jian Tang; Shanghang Zhang; Qiang Zhang; Wei Pan. Lhm-humanoid: Learning a unified policy for long-horizon humanoid whole-body loco-manipulation in diverse messy environments. *arXiv*, 2025. URL http://arxiv.org/abs/2508.16943v2.

Junha Min; Junghyeon Ma; Jiwung Kwon; Sunggyu Bae; Joohyung Kim; Kyungseo Park. Tactile-proprioceptive sensor fusion for contact wrench estimation in whole-body physical human-robot interaction. *arXiv*, 2026. URL http://arxiv.org/abs/2605.28412v1.

Kai Ploeger; Michael Lutter; Jan Peters. High acceleration reinforcement learning for real-world juggling with binary rewards. *arXiv*, 2020. URL http://arxiv.org/abs/2010.13483v3.

Michael Burke; Katie Lu; Daniel Angelov; Arturas Straizys; Craig Innes; Kartic Subr; Subramanian Ramamoorthy. Learning rewards from exploratory demonstrations using probabilistic temporal ranking. *Autonomous Robots*, 2023. doi: 10.1007/s10514-023-10120-w. URL https://doi.org/10.1007/s10514-023-10120-w.

Jaime E. Duarte; David J. Reinkensmeyer. Effects of robotically modulating kinematic variability on motor skill learning and motivation. *Journal of Neurophysiology*, 2015. doi: 10.1152/jn.00163.2014. URL https://doi.org/10.1152/jn.00163.2014.

Kayla Matheus; Rebecca Ramnauth; Brian Scassellati; Nicole Salomons. Long-term interactions with social robots: Trends, insights, and recommendations. *ACM Transactions on Human-Robot Interaction*, 2025. doi: 10.1145/3729539. URL https://doi.org/10.1145/3729539.

Jan Peters; Stefan Schaal. Reinforcement learning by reward-weighted regression for operational space control. 2007. doi: 10.1145/1273496.1273590. URL https://doi.org/10.1145/1273496.1273590.

Kai Gao; Fan Wang; Erica Aduh; Dylan Randle; Jane Shi. Must: Multi-head skill transformer for long-horizon dexterous manipulation with skill progress. *arXiv*, 2025a. URL http://arxiv.org/abs/2502.02753v1.

Kai Gao; Fan Wang; Erica Aduh; Dylan Randle; Jane Shi. Must: Multi-head skill transformer for long-horizon dexterous manipulation with skill progress. *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025b. doi: 10.1109/icra55743.2025.11127280. URL https://doi.org/10.1109/icra55743.2025.11127280.

Chenyang Zhao; Timothy Hospedales; Freek Stulp; Olivier Sigaud. Tensor based knowledge transfer across skill categories for robot control. 2017. doi: 10.24963/ijcai.2017/484. URL https://doi.org/10.24963/ijcai.2017/484.

Joao Bernardo Alves; Nuno Lau; Filipe Silva. Curriculum-guided skill learning for long-horizon robot manipulation tasks. *Robotics and Autonomous Systems*, 2025. doi: 10.1016/j.robot.2025.105032. URL https://doi.org/10.1016/j.robot.2025.105032.

Markus Knauer; Samuel Bustamante; Thomas Eiband; Alin Albu-Schaffer; Freek Stulp; Joao Silverio. Irosa: Interactive robot skill adaptation using natural language. *arXiv*, 2026. doi: 10.1109/lra.2026.3671560. URL http://arxiv.org/abs/2603.03897v3.

Lihang Feng; Long Zhang; Shiyue Ma; Dong Wang; Aiguo Song. Dht_xlstm: A hybrid temporal-mapping framework for skill-oriented grasp prediction in teleoperated robots via dual-quaternion control and context-aware learning. *Biomimetic Intelligence and Robotics*, 2026. doi: 10.1016/j.birob.2026.100296. URL https://doi.org/10.1016/j.birob.2026.100296.

Sambad Regmi; Devin Burns; Yun Seong Song. Humans modulate arm stiffness to facilitate motor communication during overground physical human-robot interaction. *Scientific Reports*, 2022. doi: 10.1038/s41598-022-23496-z. URL https://doi.org/10.1038/s41598-022-23496-z.

Xiucai Huang; Shifeng Chen; Yongduan Song. Leskill: Structured skill learning for long-horizon robotic manipulation tasks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2025. doi: 10.1109/tsmc.2025.3605528. URL https://doi.org/10.1109/tsmc.2025.3605528.

Chenguang Yang; Chao Zeng; Peidong Liang; Zhijun Li; Ruifeng Li; ChunYi Su. Interface design of a physical humanrobot interaction system for human impedance adaptive skill transfer. *IEEE Transactions on Automation Science and Engineering*, 2017. doi: 10.1109/tase.2017.2743000. URL https://doi.org/10.1109/tase.2017.2743000.

Chenyang Ran; Jianbo Su. Task-oriented self-imitation learning for robotic autonomous skill acquisition. *International Journal of Humanoid Robotics*, 2024. doi: 10.1142/s0219843624500014. URL https://doi.org/10.1142/s0219843624500014.

Yixuan Zhou; Naisheng Tang; Ziyi Li; Hanlei Sun. Methodology for humanrobot collaborative assembly based on human skill imitation and learning. *Machines*, 2025. doi: 10.3390/machines13050431. URL https://doi.org/10.3390/machines13050431.

Suguru Arimoto; Masahiro Sekimoto; Kenji Tahara. Iterative learning without reinforcement or reward for multijoint movements: A revisit of bernstein's dof problem on dexterity. *Journal of Robotics*, 2010. doi: 10.1155/2010/217867. URL https://doi.org/10.1155/2010/217867.

Mingyu Cai; Cristian-Ioan Vasile. Safety-critical learning of robot control with temporal logic specifications. *arXiv (Cornell University)*, 2021. doi: 10.48550/arxiv.2109.02791. URL https://doi.org/10.48550/arxiv.2109.02791.

Kevin Jatin Vora. Reward adaptation via q-manipulation: Provably beneficial reward function transfer in reinforcement learning. *Proceedings of the Thirty-ThirdInternational Joint Conference on Artificial Intelligence*, 2024. doi: 10.24963/ijcai.2024/1244. URL https://doi.org/10.24963/ijcai.2024/1244.

Kevin Jatin Vora. Reward adaptation via q-manipulation: Provably beneficial reward function transfer in reinforcement learning. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, 2025. doi: 10.24963/ijcai.2025/1244. URL https://doi.org/10.24963/ijcai.2025/1244.

Mythra V. Balakuntala; Upinder Kaur; Xin Ma; Juan Wachs; Richard M. Voyles. Learning multimodal contact-rich skills from demonstrations without reward engineering. *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021. doi: 10.1109/icra48506.2021.9561734. URL https://doi.org/10.1109/icra48506.2021.9561734.

Guangming Wang; Minjian Xin; Wenhua Wu; Zhe Liu; Hesheng Wang. Learning of long-horizon sparse-reward robotic manipulator tasks with base controllers. *IEEE Transactions on Neural Networks and Learning Systems*, 2022. doi: 10.1109/tnnls.2022.3201705. URL https://doi.org/10.1109/tnnls.2022.3201705.

Yu Huang; Ziji Wu; Zhengyi Ma; Kexin Ma; Ji Wang. Neuro-symbolic hierarchical learning for long-horizon robotic tasks. *Proceedings of the ACM on Programming Languages*, 2026. doi: 10.1145/3808338. URL https://doi.org/10.1145/3808338.

Yuxuan Kuang; Haoran Geng; Amine Elhafsi; Tan-Dzung Do; Pieter Abbeel; Jitendra Malik; Marco Pavone; Yue Wang. Skillblender: Towards versatile humanoid whole-body loco-manipulation via skill blending. *arXiv*, 2025. URL http://arxiv.org/abs/2506.09366v1.

Xukun Liu; Fengjuan Xie; Shibo Liu; Xu Sun; Zhenyu Liu; Guangning Li; Shenggang Wei. St-hadp: Spatio-temporal hierarchical attention diffusion policy for long-horizon generalizable bimanual visuomotor imitation. *Frontiers in Neurorobotics*, 2026. doi: 10.3389/fnbot.2026.1846433. URL https://doi.org/10.3389/fnbot.2026.1846433.

Wenhao Yu; Nimrod Gileadi; Chuyuan Fu; Sean Kirmani; Kuang-Huei Lee; Montse Gonzalez Arenas; Hao-Tien Lewis Chiang; Tom Erez; Leonard Hasenclever; Jan Humplik; Brian Ichter; Ted Xiao; Peng Xu; Andy Zeng; Tingnan Zhang; Nicolas Heess; Dorsa Sadigh; Jie Tan; Yuval Tassa; Fei Xia. Language to rewards for robotic skill synthesis. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2306.08647. URL https://doi.org/10.48550/arxiv.2306.08647.

Yiting Chen; Kenneth Kimble; Edward H. Adelson; Tamim Asfour; Podshara Chanrungmaneekul; Sachin Chitta; Yash Chitambar; Ziyang Chen; Ken Goldberg; Danica Kragic; Hui Li; Xiang Li; Yunzhu Li; Aaron Prather; Nancy Pollard; Maximo A. Roa-Garzon; Robert Seney; Shuo Sha; Shihefeng Wang; Yu Xiang. Manipulationnet: An infrastructure for benchmarking real-world robot manipulation with physical skill challenges and embodied multimodal reasoning. *arXiv*, 2026. URL http://arxiv.org/abs/2603.04363v1.

Linji Wang; Tong Xu; Yuanjie Lu; Xuesu Xiao. Reward training wheels: Adaptive auxiliary rewards for robotics reinforcement learning. *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. doi: 10.1109/iros60139.2025.11247039. URL `https://doi.org/10.1109/iros60139.2025.11247039`.

Jiayu Zhou; Qiwei Wu; Haitao Jiang; Xuanbao Qin; Yunjiang Lou; Xiaogang Xiong; Renjing Xu. Gentle manipulation of long-horizon tasks without human demonstrations. *IEEE Robotics and Automation Letters*, 2026. doi: 10.1109/lra.2026.3653406. URL `https://doi.org/10.1109/lra.2026.3653406`.

Shuo Cheng; Danfei Xu. League: Guided skill learning and abstraction for long-horizon manipulation. *IEEE Robotics and Automation Letters*, 2023a. doi: 10.1109/lra.2023.3308061. URL `https://doi.org/10.1109/lra.2023.3308061`.

Tao Huang; Guangqi Jiang; Yanjie Ze; Huazhe Xu. Diffusion reward: Learning rewards via conditional video diffusion. *arXiv*, 2023b. URL `http://arxiv.org/abs/2312.14134v3`.

Utkarsh A. Mishra; Shangjie Xue; Yongxin Chen; Danfei Xu. Generative skill chaining: Long-horizon skill planning with diffusion models. *arXiv (Cornell University)*, 2023c. doi: 10.48550/arxiv.2401.03360. URL `https://doi.org/10.48550/arxiv.2401.03360`.

Ran Cao; Long Cheng; Wenchao Xue. Iterative assist-as-needed control with skill learning for physical human-robot interaction. *IFAC-PapersOnLine*, 2023. doi: 10.1016/j.ifacol.2023.10.557. URL `https://doi.org/10.1016/j.ifacol.2023.10.557`.

Benoit Boulet; Yuwei Fu; Di Wu; Wei Xu; Haichao Zhang. Robot policy learning with temporal optimal transport reward. *Advances in Neural Information Processing Systems 37*, 2024. doi: 10.52202/079017-3879. URL `https://doi.org/10.52202/079017-3879`.

Xipei Ren; Zhifan Guo; Aobo Huang; Yuying Li; Xinyi Xu; Xiaoyu Zhang. Effects of social robotics in promoting physical activity in the shared workspace. *Sustainability*, 2022. doi: 10.3390/su14074006. URL `https://doi.org/10.3390/su14074006`.

Zhihao Li; Zhenglong Sun; Jionglong Su; Jiaming Zhang. Learning a skill-sequence-dependent policy for long-horizon manipulation tasks. *2021 IEEE 17th International Conference on Automation Science and Engineering (CASE)*, 2021. doi: 10.1109/case49439.2021.9551399. URL `https://doi.org/10.1109/case49439.2021.9551399`.

Yunhai Han; Kelin Yu; Rahul Batra; Nathan Boyd; C. H. Mehta; Tuo Zhao; Yu She; Seth Hutchinson; Ye Zhao. Learning generalizable vision-tactile robotic grasping strategy for deformable objects via transformer. *IEEE/ASME Transactions on Mechatronics*, 2024. doi: 10.1109/tmech.2024.3400789. URL `https://doi.org/10.1109/tmech.2024.3400789`.

Matthew Acs; Xiangnan Zhong. Raes: a reward-aligned expert sequencing framework for long-horizon robotics. *International Journal of Intelligent Robotics and Applications*, 2026. doi: 10.1007/s41315-026-00524-z. URL `https://doi.org/10.1007/s41315-026-00524-z`.

Xinzheng Zhang; Jianfen Zhang; Junpei Zhong. Skill learning for intelligent robot by perception-action integration: A view from hierarchical temporal memory. *Complexity*, 2017. doi: 10.1155/2017/7948684. URL `https://doi.org/10.1155/2017/7948684`.

Hongmin Wu; Wu Yan; Zhihao Xu; Taobo Cheng; Xuefeng Zhou. A framework of robot skill learning from complex and long-horizon tasks. *IEEE Transactions on Automation Science and Engineering*, 2021. doi: 10.1109/tase.2021.3127574. URL `https://doi.org/10.1109/tase.2021.3127574`.

Youchao Zhang; Fanghao Wang; Tong Zhou; Xiangyu Guo; Guang Chen; Alois Knoll; Yibin Ying; Mingchuan Zhou. Skill information representation imitation learning for long-horizon dexterous robot micromanipulation of deformable cell. *IEEE Transactions on Cybernetics*, 2026. doi: 10.1109/tcyb.2026.3656969. URL `https://doi.org/10.1109/tcyb.2026.3656969`.

Borui Nan; Xinyi Zheng; Wanru Gong; Lanyue Bi; Xiaoyv Zhu; Xiaoqing Zhu. Robot motion skill learning method based on focused reward transformer. *Scientific Reports*, 2025. doi: 10.1038/s41598-025-95881-3. URL `https://doi.org/10.1038/s41598-025-95881-3`.

Lingxuan Wu; Zijian Zhu; Lizhong Wang; Chengyang Ying; Huayu Chen; Xiao Yang; Fangming Liu; Jun Zhu. Pact: Self-evolving physical safety alignment for diffusion policies in embodied manipulation. *arXiv*, 2026. URL http://arxiv.org/abs/2606.08414v1.

Yifeng Zhu; Peter Stone; Yuke Zhu. Bottom-up skill discovery from unsegmented demonstrations for long-horizon robot manipulation. *IEEE Robotics and Automation Letters*, 2022. doi: 10.1109/lra.2022.3146589. URL https://doi.org/10.1109/lra.2022.3146589.